

JEDEC PUBLICATION

JC-42.6 Manufacturer Identification (ID) Code for Low Power Memories

JEP166F

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JEDEC SOLID STATE TECHNOLOGY ASSOCIATION



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JC-42.6 Manufacturer Identification (ID) Code for Low Power Memories

(From JEDEC Board Ballot JCB-12-64, JCB-14-48, and JCB-15-47, formulated under the cognizance of the JC-42.6 Subcommittee on Low Power Memory.)

1 Scope

This document defines the JC-42.6 Manufacturer ID. This document covers Manufacturer ID Codes for the following technologies: LPDDR (JESD209), LPDDR2 (JESD209-2), LPDDR3 (JESD209-3), LPDDR4 (JESD209-4), LPDDR5 (JESD209-5), LPDDR6(JESD209-6), Wide-IO (JESD229), and Wide-IO2 (JESD229-2). The purpose of this document is to define the Manufacturer ID for these devices.

2 LPDDR (JESD209) Manufacturer ID

LPDDR Manufacturer ID			
DQ3	DQ2	DQ1	DQ0
LPDDR Manufacturer ID	Read-Only	OP<3:0>	0000B : Reserved
			0001B : Samsung
			0010B : Infineon
			0011B : Elpida
			0100B : Etron
			0101B : Nanya
			0110B : SK hynix
			0111B : Mosel
			1000B : Winbond
			1001B : ESMT
			1010B : NVM
			1011B : ISSI
			1100B : JSC
			1101B : AP Memory
			1110B : all others
			1111B : Micron

3 LPDDR2 and LPDDR3 (JESD209-2 and JESD209-3) Manufacturer ID

LPDDR2/LPDDR3 Manufacturer ID							
OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
LPDDR2/LPDDR3 Manufacturer ID	Read-Only	OP<7:0>	0000 0000B : Reserved				
			0000 0001B : Samsung				
			0000 0010B : Qimonda				
			0000 0011B : Elpida				
			0000 0100B : Etron				
			0000 0101B : Nanya				
			0000 0110B : SK hynix				
			0000 0111B : Mosel				
			0000 1000B : Winbond				
			0000 1001B : ESMT				
			0000 1010B : Zentel				
			0000 1011B : Spansion				
			0000 1100B : SST				
			0000 1101B : ZMOS				
			0000 1110B : Intel				
			0000 1111B : Reserved				
			0001 0000B : Reserved				
			0001 0001B : Reserved				
			0001 0010B : Being Advanced Memory Corp				
			0001 0011B : Reserved				
			0001 1010B : Xi'an UniIC Semiconductors Co., Ltd				
			0001 0100B : UNIM Innovation(Wuxi) Co., Ltd				
			0001 1011B : ISSI				
			0001 1100B : JSC				
			1010 1010B : Tezzaron				
			1100 0010B : Macronix				
			1100 0100B : HT Micron Semicondutores S.A				
			1100 0101B : SINKER				
			1111 1000B : Fidelix				
			1111 1100B : eveRAM				
			1111 1101B : AP Memory				
			1111 1110B : Numonyx				
			1111 1111B : Micron				
			All Others : Reserved				

4 WIDE I/O (JESD229) Manufacturer ID

Wide I/O Manufacturer ID			
DQ3	DQ2	DQ1	DQ0
Wide I/O Manufacturer ID			
Wide I/O Manufacturer ID	Read-Only	OP<3:0>	0000B : Reserved 0001B : Samsung 0010B : Being Advanced Memory Corporation 0011B : Elpida 0100B : Reserved 0101B : Nanya 0110B : SK hynix 0111B : Reserved 1000B : Winbond 1001B : ESMT 1010B : Reserved 1011B : Reserved 1100B : Reserved 1101B : Reserved 1110B : Reserved 1111B : Micron



5 LPDDR4 (JESD209-4) Manufacturer ID

LPDDR4 Manufacturer ID							
OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
LPDDR4 Manufacturer ID	Read-Only	OP<7:0>	0000 0000B : Reserved				
			0000 0001B : Samsung				
			0000 0010B : Reserved*				
			0000 0011B : Reserved*				
			0000 0100B : Etron				
			0000 0101B : Nanya				
			0000 0110B : SK hynix				
			0000 0111B : Reserved*				
			0000 1000B : Winbond				
			0000 1001B : ESMT				
			0000 1010B : Reserved				
			0000 1011B : Reserved*				
			0000 1100B : Reserved*				
			0000 1101B : Reserved*				
			0000 1110B : Reserved*				
			0000 1111B : Reserved*				
			0001 0010B : Reserved*				
			0001 0011B : CXMT				
			0001 1010B : Xi'an UniIC Semiconductors Co., Ltd				
			0001 0100B : UNIM Innovation (Wuxi) Co., Ltd				
			0001 1011B : ISSI				
			0001 1100B : JSC				
			1001 1100B : Shenzhen Wodposit Electronics				
			1010 1010B : Reserved*				
			1100 0010B : Reserved*				
			1100 0100B : Reserved*				
			1100 0101B : SINKER				
			1110 0101B : Dosilicon Co,Ltd.				
			1111 1000B : Fidelix				
			1111 1001B : Ultra Memory				
			1111 1100B : Reserved*				
			1111 1101B : AP Memory				
			1111 1110B : Reserved*				
			1111 1111B : Micron				
			All others : Reserved				

* Reserved codes highlighted with an asterisk (*) have vendor name assignments in other standards in this document. Use caution when assigning those codes in the future.

6 LPDDR5 (JESD209-5) Manufacturer ID

LPDDR5 Manufacturer ID							
OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
LPDDR5 Manufacturer ID	Read-Only	OP<7:0>	0000 0000B : Reserved				
			0000 0001B : Samsung				
			0000 0010B : Reserved*				
			0000 0011B : Reserved*				
			0000 0100B : Etron				
			0000 0101B : Nanya				
			0000 0110B : SK hynix				
			0000 0111B : Reserved*				
			0000 1000B : Winbond				
			0000 1001B : ESMT				
			0000 1010B : Reserved				
			0000 1011B : Reserved*				
			0000 1100B : Reserved*				
			0000 1101B : Reserved*				
			0000 1110B : Reserved*				
			0000 1111B : Reserved*				
			0001 0010B : Reserved*				
			0001 0011B : CXMT				
			0001 0100B : Reserved*				
			0001 1010B : Reserved				
			0001 1011B : Reserved*				
			0001 1100B : Reserved				
			1000 1111B : CXSH Co. Ltd.				
			1001 1011B : YMTC				
			1001 1100B : Shenzhen Wodposit Electronics				
			1010 1010B : Reserved*				
			1100 0010B : Reserved*				
			1100 0100B : Reserved*				
			1100 0101B : Reserved*				
			1100 1101B : Piecemakers Technology				
			1110 0101B : Dosilicon Co, Ltd.				
			1111 1000B : Reserved				
			1111 1001B : Reserved				
			1111 1100B : Reserved*				
			1111 1101B : Reserved				
			1111 1110B : Reserved*				
			1111 1111B : Micron				
			All others : Reserved				

* Reserved codes highlighted with an asterisk (*) have vendor name assignments in other standards in this document. Use caution when assigning those codes in the future.

7 LPDDR6 (JESD209-6) Manufacturer ID

LPDDR6 Manufacturer ID							
OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
LPDDR6 Manufacturer ID	Read-Only	OP<7:0>	0000 0000B : Reserved				
			0000 0001B : Samsung				
			0000 0010B : Reserved*				
			0000 0011B : Reserved*				
			0000 0100B : Reserved*				
			0000 0101B : Nanya				
			0000 0110B : SK hynix				
			0000 0111B : Reserved*				
			0000 1000B : Winbond				
			0000 1001B : ESMT				
			0000 1010B : Reserved				
			0000 1011B : Reserved*				
			0000 1100B : Reserved*				
			0000 1101B : Reserved*				
			0000 1110B : Reserved*				
			0000 1111B : Reserved*				
			0001 0010B : Reserved*				
			0001 0011B : CXMT				
			0001 0100B : Reserved*				
			0001 1010B : Reserved				
			0001 1011B : Reserved*				
			0001 1100B : Reserved				
			1001 1011B : YMTc				
			1001 1100B : Reserved*				
			1010 1010B : Reserved*				
			1100 0010B : Reserved*				
			1100 0100B : Reserved*				
			1100 0101B : Reserved*				
			1110 0101B : Reserved*				
			1111 1000B : Reserved				
			1111 1001B : Reserved				
			1111 1100B : Reserved*				
			1111 1101B : Reserved				
			1111 1110B : Reserved*				
			1111 1111B : Micron				
			All others : Reserved				

* Reserved codes highlighted with an asterisk (*) have vendor name assignments in other standards in this document. Use caution when assigning those codes in the future.

8 Summary of LPDDR2/3/4/5/6 Manufacturer ID

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
LPDDR2/3/4/5/6 Manufacturer ID							

LPDDR	LPDDR2/3	LPDDR4	LPDDR5	LPDDR6
0000B : Reserved	0000 0000B : Reserved	0000 0000B : Reserved	0000 0000B : Reserved	0000 0000B : Reserved
0001B : Samsung	0000 0001B : Samsung	0000 0001B : Samsung	0000 0001B : Samsung	0000 0001B : Samsung
0010B : Infineon	0000 0010B : Qimonda	0000 0010B : Reserved*	0000 0010B : Reserved*	0000 0010B : Reserved*
0011B : Elpida	0000 0011B : Elpida	0000 0011B : Reserved*	0000 0011B : Reserved*	0000 0011B : Reserved*
0100B : Etron	0000 0100B : Etron	0000 0100B : Etron	0000 0100B : Etron	0000 0100B : Reserved*
0101B : Nanya	0000 0101B : Nanya	0000 0101B : Nanya	0000 0101B : Nanya	0000 0101B : Nanya
0101B : SKHynix	0000 0110B : SK hynix	0000 0110B : SK hynix	0000 0110B : SK hynix	0000 0110B : SK hynix
0111B : Mosel	0000 0111B : Mosel	0000 0111B : Reserved*	0000 0111B : Reserved*	0000 0111B : Reserved*
0111B : Winbond	0000 1000B : Winbond	0000 1000B : Winbond	0000 1000B : Winbond	0000 1000B : Winbond
0111B : ESMT	0000 1001B : ESMT	0000 1001B : ESMT	0000 1001B : ESMT	0000 1001B : ESMT
1010B : NVM	0000 1010B : Zentel	0000 1010B : Reserved*	0000 1010B : Reserved*	0000 1010B : Reserved*
1011B : ISSI	0000 1011B : Spansion	0000 1011B : Reserved*	0000 1011B : Reserved*	0000 1011B : Reserved*
1100B : JSC	0000 1100B : SST	0000 1100B : Reserved*	0000 1100B : Reserved*	0000 1100B : Reserved*
1101B : AP Memory	0000 1101B : ZMOS	0000 1101B : Reserved*	0000 1101B : Reserved*	0000 1101B : Reserved*
1110B : all others	0000 1110B : Intel	0000 1110B : Reserved*	0000 1110B : Reserved*	0000 1110B : Reserved*
1111B : Micron	0000 1111B : Reserved* Be careful, could conflict with Micron LPDDR code	0000 1111B : Reserved* Be careful, could conflict with Micron LPDDR code	0000 1111B : Reserved* Be careful, could conflict with Micron LPDDR code	0000 1111B : Reserved* Be careful, could conflict with Micron LPDDR code
	0001 0010B : Being Advanced Memory Corp	0001 0010B : Reserved*	0001 0010B : Reserved*	0001 0010B : Reserved*
	0001 0011B : Reserved	0001 0011B : CXMT	0001 0011B : CXMT	0001 0011B : CXMT
	0001 0100B : UNIM Innovation (Wuxi) Co., Ltd	0001 0100B : UNIM Innovation (Wuxi) Co., Ltd	0001 0100B : Reserved	0001 0100B : Reserved
	0001 1010B : Xi'an UnilC Semiconductors Co., Ltd	0001 1010B : Xi'an UnilC Semiconductors Co., Ltd	0001 1010B : Reserved	0001 1010B : Reserved
	0001 1011B : ISSI	0001 1011B : ISSI	0001 1011B : Reserved*	0001 1011B : Reserved*
	0001 1100B : JSC	0001 1100B : JSC	0001 1100B : Reserved	0001 1100B : Reserved
	1000 1111B : Reserved	1000 1111B : Reserved	1000 1111B : CXSH Co. Ltd.	1000 1111B : Reserved
	1001 1011B : Reserved	1001 1011B : Reserved	1001 1011B : YMTC	1001 1011B : YMTC
	1001 1100B : Reserved	1001 1100B : Shenzhen Wodposit Electronics	1001 1100B : Shenzhen Wodposit Electronics	1001 1100B : Reserved
	1010 1010B : Tezzaron	1010 1010B : Reserved*	1010 1010B : Reserved*	1010 1010B : Reserved*

	1100 0010B : Macronix	1100 0010B : Reserved*	1100 0010B : Reserved*	1100 0010B : Reserved*
	1100 0100B : HT Micron Semicondutores S.A.	1100 0100B : Reserved*	1100 0100B : Reserved*	1100 0100B : Reserved*
	1100 0101B : Reserved*	1100 0101B : SINKER	1100 0101B : Reserved*	1100 0101B : Reserved*
	1110 0101B : Dosilicon Co,Ltd.	1110 0101B : Reserved*	1110 0101B : Dosilicon Co,Ltd.	1110 0101B : Reserved*
	1111 1000B : Fidelix	1111 1000B : Fidelix	1111 1000B : Reserved	1111 1000B : Reserved
	1111 1001B : Reserved	1111 1001B : Ultra Memory	1111 1001B : Reserved	1111 1001B : Reserved
	1111 1100B : eveRAM	1111 1100B : Reserved*	1111 1100B : Reserved*	1111 1100B : Reserved*
	1111 1101B : AP Memory	1111 1101B : AP Memory	1111 1101B : Reserved	1111 1101B : Reserved
	1111 1110B : Numonyx	1111 1110B : Reserved*	1111 1110B : Reserved*	1111 1110B : Reserved*
	1111 1111B : Micron	1111 1111B : Micron	1111 1111B : Micron	1111 1111B : Micron
	All others : Reserved	All others : Reserved	All others : Reserved	All others : Reserved



9 WIDE I/O2 (JESD229-2) Manufacturer ID

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Wide I/O2 Manufacturer ID							
Wide I/O 2 Manufacturer ID	Read- Only	OP<7:0>	0000 0000B : Reserved				
			0000 0001B : Samsung				
			0000 0010B : Reserved*				
			0000 0011B : Reserved*				
			0000 0100B : Reserved*				
			0000 0101B : Nanya				
			0000 0110B : SK hynix				
			0000 0111B : Reserved*				
			0000 1000B : Winbond				
			0000 1001B : ESMT				
			0000 1010B : Reserved				
			0000 1011B : Reserved*				
			0000 1100B : Reserved*				
			0000 1101B : Reserved*				
			0000 1110B : Reserved*				
			0000 1111B : Reserved*				
			0001 0010B : Reserved*				
			0001 1011B : Reserved*				
			0001 1100B : Reserved*				
			1100 0010B : Reserved*				
			1111 1110B : Reserved*				
			1111 1111B : Micron				
			All others : Reserved				

* Reserved codes highlighted with an asterisk (*) have vendor name assignments in other standards in this document. Use caution when assigning those codes in the future.

Annex A (informative) Obtaining a JC-42.6 Manufacturer Identification Code (ID)

JC-42.6 Manufacturer Identification Codes for Low-Power Memories are assigned, maintained, and updated by the JC-42.6 Subcommittee. The JC-42.6 chairperson is responsible for assigning the ID code. The JEDEC office will update JEP166 editorially and publish it in a timely manner.

A new Manufacturer Identification Code for Low-Power Memories can be obtained at the following web page <https://www.jedec.org/id-codes-low-power-memories>.



Annex B (informative) Differences between revisions

This annex briefly describes most of the additions/changes made to entries that appear in this publication, JEP166F, compared to its predecessors.

Clause	Description of change
1	Added LPDDR6 to scope
3,5,6	New ID codes assigned
7	New clause for LPDDR6
8	The former clause 7 is now clause 8
9	The former clause 8 is now clause 9

B.1 Differences between JEP166D and JEP166E (July 2023)

Clause	Description of change
All	Added “xxx” to 0001 0xxx011B.
3, 5, 6	New ID codes assigned
6	The former clause 6 is now clause 8
7	New clause added

B.2 Differences between JEP166C and JEP166B (February 2016)

Clause	Description of change
5	Added “CXMT” to 0001 0011B.

B.3 Differences between JEP166C and JEP166B (February 2016)

Clause	Description of change
2, 3 and 5	“EMLSI” changed to “JSC” (LPDDR4)
3, 5	Added: “0001 1010B : Xi'an UnilC Semiconductors Co., Ltd” (LPDDR4 and LPDDR2/3)
5	Added: “1111 1001B : Ultra Memory” (LPDDR4)

B.4 Differences between JEP166B and JEP166A (December 2014)

Clause	Description of change
Annex A	modified with new information
2, 3, 5	new ID codes assigned

B.5 Differences between JEP166A and JEP166 (March 2014)

Clause	Description of change
1	In Scope: added reference to JESD209-4, and JESD229-2
5	Added
6	Added



Standard Improvement Form

JEDEC **JEP166F**

The purpose of this form is to provide the Technical Committees of JEDEC with input from the industry regarding usage of the subject standard. Individuals or companies are invited to submit comments to JEDEC. All comments will be collected and dispersed to the appropriate committee(s).

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1. I recommend changes to the following:

☐ Requirement, clause number _____

☐ Test method number _____ Clause number _____

The referenced clause number has proven to be:

☐ Unclear ☐ Too Rigid ☐ In Error

☐ Other _____

2. Recommendations for correction:

3. Other suggestions for document improvement:

Submitted by

Name: _____

Phone: _____

Company: _____

E-mail: _____

Address: _____

City/State/Zip: _____

Date: _____

